

RFID Technology for Building Access Control

Access control is a common application of RFID technology. Passive RFID tags are an ideal candidate for this application as these have a very short read range of 10-20 cm only



T.K. HAREENDRAN

Numerous radio-frequency identification (RFID) applications are implemented in various sectors for very different purposes. Conventional examples include automatic electronic tracking of goods in supply chains, cashless payments and physical building access with an electronic card.

Non-traditional applications apply

the ubiquitous and permeative nature of RFID in new ways. Specifically, RFID is being used as an interface embedded throughout the environment for data collection or for seamlessly calling for services and information. Recent RFID applications exploit the inexpensive and miniature form factor of RFID tags to sync real and virtual worlds.

RFID readers collect data wirelessly from electronic transponders, also called tags, which are attached to or embedded in objects. RFID systems are built around software, networks and databases that enable information to flow from tags to the infrastructure where it is processed and stored.

RFID systems are created to suit the intended application. Some use low-cost passive tags, which have low data capacity and

also short read range. Most data is put on the network, and only a small amount of information is loaded onto tags. Others use sophisticated, high-performance tags with high read ranges. These may store considerable data on tags, without network connection.

Roots of RFID

RFID has been dubbed the world's oldest new technology. It dates back

to the 19th century with applications related to friend or foe identification of military aircrafts. The 'identify friend or foe' (IFF) system allowed radar operators and pilots to automatically distinguish a friendly aircraft from enemies' via radio-frequency signals. IFF systems helped prevent 'friendly fire' incidents and assisted in intercepting the enemy aircrafts. The first commercial applications appeared in the 1960s in the area of electronic article surveillance to fight against product theft—an application that is still widely used today.

Automatic object detection systems preceded automatic RFID systems. According to the *Handbook of Technology Management*, one of the earliest patented systems was a radio transmitter for object detection system designed by John Logie Baird in 1926. The passive communication technology often used in RFID was first presented in Henry Stockman's seminal

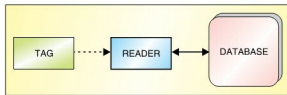


Fig. 1: RFID system interaction

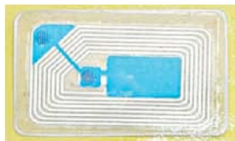


Fig. 2: RFID proximity card